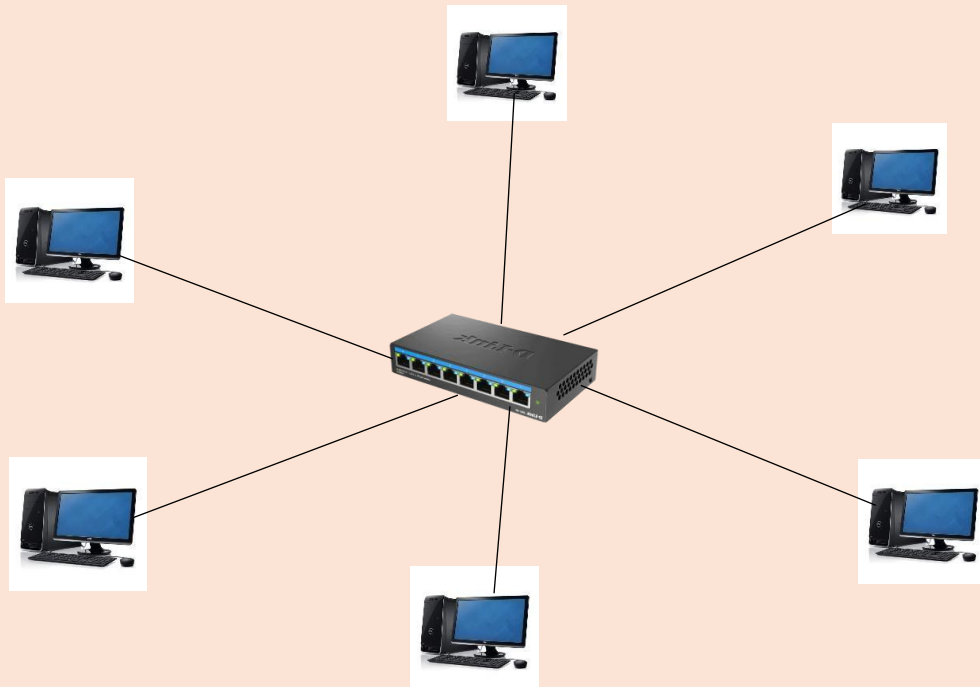


Network topology

1. Draw and label diagrams of star, bus, and mesh network topologies on paper or using any drawing tool. For each, write one real-world app or service (like WhatsApp, Flipkart, or IPL streaming) that could use that topology and explain why in 1-2 lines.



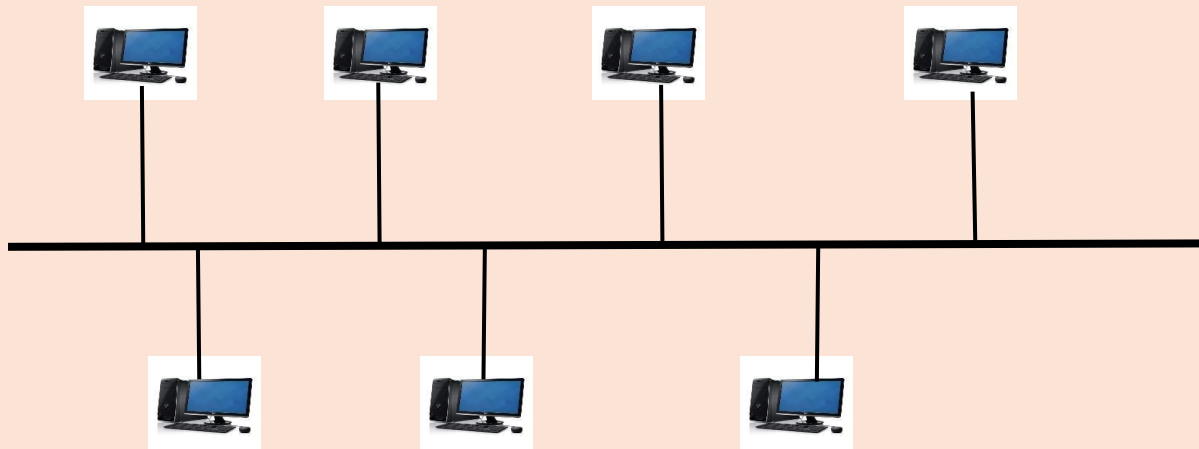
Star Topology



Ex :- Home Wi-Fi Network

A home Wi-Fi network is a real-world example of a star topology because all devices connect to one central router. It is used because it is reliable, easy to manage, and a problem with one device does not affect the others.

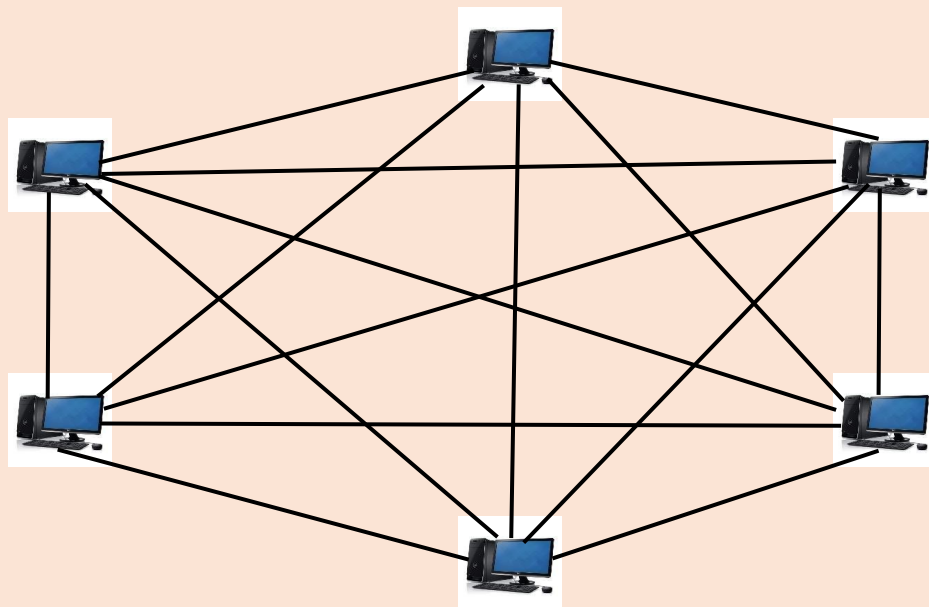
Bus topology



Ex:- A small office where a few computers are connected using a single Ethernet cable.

It allows multiple computers to communicate and share resources using one main cable, making installation simple and inexpensive.

Mesh topology



Ex: Google Data Center

Use: Servers are connected with multiple paths. If one connection fails, data automatically uses another path, ensuring the service stays online without interruption.

A mesh topology connects every device to multiple other devices, providing high reliability, fault tolerance, and continuous communication even if one connection fails.

2. Create a comparison table listing at least four pros and four cons for each of these topologies: star, bus, ring, mesh. Include a column for 'Best Use Case' and give an example from apps you use (e.g., Instagram, Zomato, Paytm).



pros

1. Star topology is easy to install and manage.
2. If one device fails, the other devices continue to work normally.
3. It is easy to find and fix network problems.
4. New devices can be added without affecting the existing network.

Cons

1. It requires more cables than other topologies.
2. It depends on a central hub or switch.
3. If the central device fails, the entire network stops working.
4. Installation cost is higher because of the extra hardware.

Best Use Case: Home, school, office, and Wi-Fi networks.

Example: Instagram uses a star-like network where users connect to a central server through the internet.

2. Bus Topology

Pros

1. Bus topology is easy to install.
2. It uses less cable, making it cost-effective.
3. It is suitable for small networks.
4. It does not require a central networking device.

Cons

1. If the main cable fails, the whole network stops working.
2. Network performance becomes slow when many devices are connected.
3. It is difficult to find cable faults.
4. It is not suitable for large networks.

Best Use Case: Small offices or temporary networks.

Example: A small local file-sharing network in a computer lab.

3. Ring Topology

Pros

1. Data travels in one direction, reducing collisions.
2. All devices get equal access to the network.
3. It performs well under heavy network traffic.
4. It provides predictable data transfer.

Cons

1. If one device or cable fails, the entire network can stop.
2. Adding or removing devices is difficult.
3. Troubleshooting takes more time.
4. It is less flexible than a star topology.

Best Use Case: Industrial control systems and some older network systems.

Example: Some banking and telecom networks use ring topology for reliable communication.

4. Mesh Topology

Pros

1. Mesh topology is highly reliable.
2. It provides multiple paths for data transmission.
3. If one connection fails, communication continues through another path.
4. It offers excellent fault tolerance and network availability.

Cons

1. It requires a large number of cables.
2. Installation and maintenance are expensive.
3. Network setup is complex.
4. It is not practical for small networks.

Best Use Case: Data centers, military networks, and wireless mesh Wi-Fi systems.

Example: Paytm, Zomato, and Instagram servers use mesh-like connections in data centers to ensure their services remain available even if one network link fails.

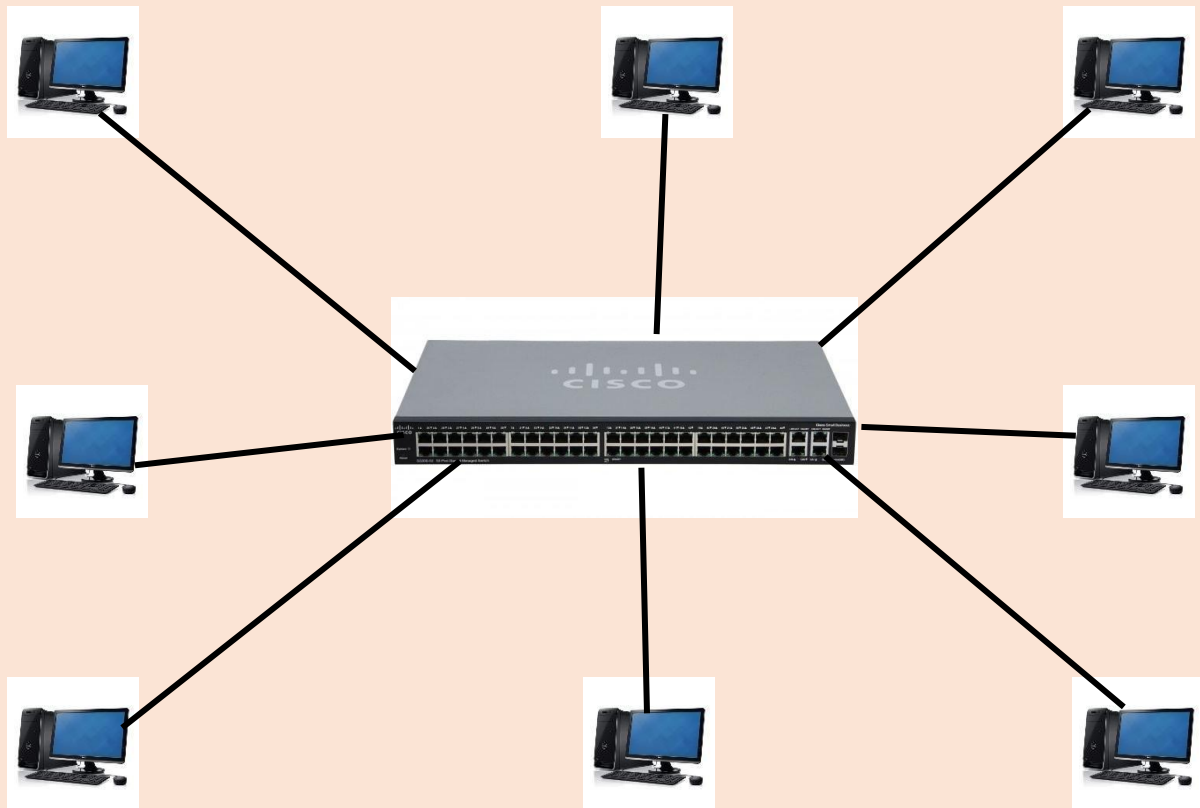
3. Imagine you are designing a new multiplayer mobile game like PUBG or Free Fire. Choose a suitable network topology for the game servers and justify your choice in 3-4 lines, mentioning how it would impact speed, reliability, and cost.

Suitable Network Topology: Mesh Topology

I would choose Mesh Topology for the game servers. It provides high speed and reliability because data can travel through multiple paths. If one server or connection fails, the game continues to run using another path, reducing downtime. Although mesh topology is expensive to set up, it is the best choice for multiplayer games like PUBG or Free Fire because it gives players a smooth and stable gaming experience.

**4. Given this scenario: A small Flipkart-style warehouse wants to connect 8 computers for order processing and inventory updates. Recommend a network topology, sketch a quick diagram, and explain your reasoning in 2-3 lines.

Hint: Consider factors like cost, ease of troubleshooting, and future expansion.**



A star topology is the best choice for a Flipkart-style warehouse because all 8 computers connect to a central switch. It is easy to install, troubleshoot, and expand by adding more computers in the future. If one computer or cable fails, the other computers continue to work normally, making it a reliable and cost-effective solution.